CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

Experiment 5: Zero-Shot, One-Shot, and Few-Shot Prompting for Promotional Social Media Post Generation

1. Aim

To design and compare Zero-shot, One-shot, and Few-shot prompts for generating a promotional social media post for an AI Workshop using Python in Google Colab.

2. Algorithm

1. Start the program.
2. Define a Zero-shot prompt that asks for an AI Workshop promotional post without examples.
3. Define a One-shot prompt that includes one example of a promotional post.
4. Define a Few-shot prompt that includes multiple examples of promotional posts.
5. Generate a sample social media post for each prompting technique.
6. Display the prompts and their corresponding outputs.
7. Compare the outputs based on creativity, clarity, tone, and completeness.
8. Display the result and stop the program.

3. Python Code (Google Colab)

Copy the following code into a Google Colab cell and click **Run**. No additional libraries or API key are required.

Python

Run

```
# Experiment 5: AI Workshop Promotional Post Generation


# Zero-shot prompt

zero_shot = """

Write an attractive social media post to promote an AI Workshop.

Include the benefits, invitation, and registration details.

"""


# One-shot prompt
```

one_shot = """

Example:

Join our Python Workshop! Learn coding, build projects,

and improve your skills. Register today!

Now write a promotional social media post for an AI Workshop.

Include benefits and a call to action.

"""

Few-shot prompt

few_shot = """

Example 1:

Explore the world of Python! Learn coding through hands-on

activities. Register now!

Example 2:

Discover Data Science! Learn data analysis and visualization

from experts. Join us today!

Now create a similar promotional social media post for

an AI Workshop. Include learning benefits and a call to action.

"""

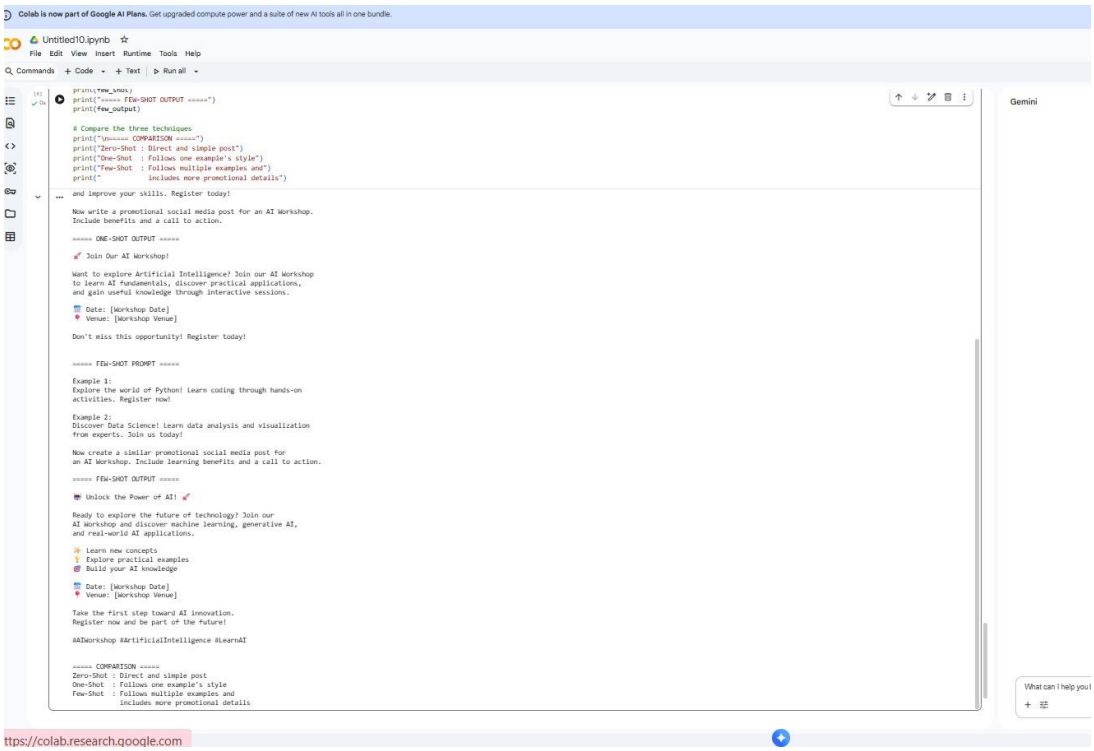
Sample outputs for comparison

zero_output = """

4. Input

- **Zero-shot:** A direct instruction to create an AI Workshop promotional post.
- **One-shot:** One example of a promotional post followed by the task.
- **Few-shot:** Two examples of promotional posts followed by the task.

5. Output



6. Comparison Table

Criteria	**Zero-shot**	**One-shot**	**Few-shot**
Creativity	Good	Good	Very good
Clarity	Good	Good	Very good
Promotional tone	Good	Good	Very good
Formatting	Basic	Good	Very good
Details	General	Moderate	More detailed

7. Result

Zero-shot, One-shot, and Few-shot prompting techniques were designed and compared for generating a promotional social media post for an AI Workshop. The Few-shot approach produced a more structured and detailed sample by following multiple examples.

Note: This beginner-friendly Colab program uses predefined sample outputs to demonstrate the three techniques. It does not call a live AI model API.